Text Mining and Sentiment Analysis Based Intelligent System for Monitoring and Evaluating Public Health Awareness Campaigns

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Abstract

The COVID-19 pandemic generated an unprecedented volume of social media discourse, creating both an opportunity and a challenge for public health authorities seeking to monitor public sentiment and counter misinformation in real time. This paper presents a unified text mining and sentiment analysis framework designed for public health awareness campaign monitoring. The system is evaluated on a large-scale dataset of 409,915 COVID-19 tweets collected across three temporal windows spanning April 2020 to June 2021. A Long Short-Term Memory (LSTM) deep learning model is trained for three-class sentiment classification — negative, neutral, and positive — achieving a test accuracy of 98.79% and a macro-average F1-score of 0.99, outperforming comparable studies in the literature. A complementary TF-IDF and Logistic Regression pipeline for misinformation detection attains 99.95% accuracy using a domain-adapted keyword lexicon. Beyond classification, the system incorporates statistical temporal spike detection based on a two-standard-deviation threshold, engagement-weighted sentiment scoring, geographic regional analysis, and a rule-based policy recommendation engine grounded in health communication principles. The entire pipeline is encapsulated within an interactive Streamlit web dashboard enabling real-time single-tweet analysis and bulk campaign monitoring. Eight negative sentiment spike dates were detected and found to align with documented pandemic events. The system correctly recommended a targeted fact-check strategy based on the corpus-level negative sentiment rate of 26.74%. This work addresses a critical gap between NLP research and deployable health communication tools.

Keywords: sentiment analysis, text mining, COVID-19, misinformation detection, LSTM, TF-IDF, public health, Twitter, Streamlit, spike detection, NLP, deep learning, infodemic

I. Introduction

The COVID-19 pandemic fundamentally altered the dynamics of global public health communication. On January 30, 2020, the World Health Organization officially declared COVID-19 a Public Health Emergency of International Concern — its highest level of alert — triggering an unprecedented global surge in public discourse [1]. Social media platforms, particularly Twitter, became primary channels through which individuals expressed anxiety, confusion, solidarity, political opinion, and harmful misinformation [1][2]. Lamsal [2] documented that COVID-19 Twitter datasets spanning April 2020 to mid-2021 captured hundreds of millions of posts reflecting rapidly shifting public attitudes, and demonstrated through longitudinal analysis that dataset design — including temporal windowing and deduplication — substantially influences the sentiment distributions observed in downstream studies. Lampos et al. [27] further established that signals extracted from online search and social media can track epidemiological trends such as infection rates and mortality with time lags of only one to two days, confirming the real-time informational value of social media monitoring for public health surveillance. The volume and velocity of pandemic-

driven data far exceeded the capacity of traditional manual surveillance, prompting urgent calls for automated computational tools capable of tracking sentiment trajectories and identifying false health claims at scale [3][26].

Sentiment analysis — the computational identification of subjective opinion, emotion, or attitude expressed in natural language — has matured significantly since its early applications to product review mining [4][26]. Pang and Lee [4] established a foundational three-level taxonomy covering document-level polarity, sentence-level subjectivity, and fine-grained aspect-level opinion mining, each with distinct technical challenges. Liu [26] subsequently characterised the key difficulties unique to social media text: informal orthography, heavy use of abbreviations and hashtags, code-switching, and the frequent absence of explicit sentiment markers. Recurrent neural architectures, in particular Long Short-Term Memory networks [5], demonstrated strong capabilities for sequential text classification by addressing the vanishing gradient problem that had limited earlier recurrent networks, enabling models to retain relevant contextual signals across the full span of a tweet. More recently, Devlin et al. [6] introduced BERT, achieving state-of-the-art results across eleven NLP benchmarks simultaneously through large-scale masked language model pre-training and task-specific fine-tuning. However, BERT's inference latency and memory footprint — typically requiring multi-GPU serving infrastructure — frequently preclude deployment in real-time, interactive systems, a practical constraint that directly motivates the LSTM-based design.

Alongside sentiment monitoring, the COVID-19 period witnessed what the World Health Organization formally termed an 'infodemic' — a flood of information, both accurate and false, making it difficult for people to find trustworthy guidance when they need it most [3]. The WHO's Infodemic Management Technical Report [3] identified social media as the primary amplification vector for pandemic misinformation and called explicitly for computational monitoring tools capable of operating at platform scale. Pulido et al. [22] quantified this dynamic on Twitter, finding that while false information spread broadly, verified science-based content was ultimately retweeted at higher rates, suggesting that credibility signals embedded in user behaviour can assist automated detection. Cinelli et al. [23] extended this analysis across Twitter, Instagram, YouTube, Reddit, and Gab, revealing that while misinformation prevalence varied by platform, echo-chamber structures amplified false content disproportionately within susceptible user communities. Alam et al. [19] operationalised detection by modelling the COVID-19 infodemic from the perspectives of journalists, fact-checkers, social media platforms, and policymakers simultaneously — demonstrating that multi-stakeholder labelling substantially improved classifier robustness compared to single-perspective annotation schemes. Automated identification of health misinformation has therefore become a research priority spanning lexicon-based detection, classical machine learning with engineered features, and neural end-to-end classifiers [19][22][23].

Despite substantial academic attention to both tasks independently, integrated systems combining sentiment analysis, misinformation detection, temporal anomaly detection, and interactive deployment remain rare. Most published studies terminate at model evaluation and do not provide tools that health practitioners can operate directly. This gap between research outputs and operational utility represents the central motivation for the present work.

This paper makes the following contributions. First, it reports state-of-the-art sentiment classification accuracy of 98.79% on a dataset of 409,915 COVID-19 tweets using an LSTM architecture, surpassing comparable work in the literature. Second, it integrates a TF-IDF-based misinformation classifier achieving 99.95% accuracy. Third, it introduces a statistical spike detection mechanism, engagement-weighted sentiment scoring, and a rule-based policy recommendation engine. Fourth, and most critically, it delivers these capabilities through a fully deployed Streamlit web dashboard accessible to non-technical public health staff. The remainder of this paper is organised as follows: Section II reviews related literature across five sub-domains; Section III describes the dataset and preprocessing pipeline; Section IV details the proposed methodology; Section V reports experimental results including three comparative tables; Section VI discusses findings, limitations, and future directions; and Section VII concludes the paper.

II. Literature Review

A. Foundations of Sentiment Analysis

The field of opinion mining and sentiment analysis was systematically surveyed by Pang and Lee [4], who established the foundational taxonomy of document-, sentence-, and aspect-level analysis and provided the first rigorous evaluation of machine learning methods — including Naive Bayes, maximum entropy, and SVM — against hand-crafted linguistic rules for review polarity classification. Their key finding was that machine learning consistently outperformed linguistic rule systems on binary polarity tasks, even when the feature engineering was linguistically motivated. Liu [26] subsequently extended this framework specifically to social media, characterising the additional difficulties introduced by informal language, emoji-heavy expression, sarcasm, and the absence of standard sentence boundaries in tweet text. Pak and Paroubek [14] were among the first to apply sentiment classification directly to Twitter, using emoticons as distant supervision labels to construct a large three-class corpus (positive, negative, neutral) without manual annotation, and showing that n-gram-based SVM classifiers achieved approximately 82% accuracy — a result that established the Twitter sentiment baseline for subsequent studies. Their construction method, exploiting emoticons as proxy labels, has since been adopted in multiple COVID-19 Twitter corpus construction pipelines [2][7]. The introduction of Word2Vec by Mikolov et al. [20] then marked a paradigm shift: dense vector representations capturing semantic similarity between terms enabled models to generalise across synonymous expressions that lexicon-based methods treated as unrelated. Pennington et al. [17] extended this through GloVe, incorporating global co-occurrence statistics to produce embeddings that better reflected corpus-level distributional patterns, particularly beneficial for low-frequency but sentiment-discriminative terms. The LSTM proposed by Hochreiter and Schmidhuber [5] completed the foundational toolkit by providing a recurrent architecture whose input, forget, and output gating mechanisms allow selective retention of long-range contextual signals — essential for tweets where sentiment cues may appear at positions far from the target word.

The introduction of word embeddings by Mikolov et al. [20] marked a paradigm shift in NLP, enabling dense vector representations that capture semantic similarity. GloVe embeddings [17] extended this approach to leverage global co-occurrence statistics. Both embedding strategies have been incorporated into sentiment classification pipelines, typically as initialisation for LSTM or CNN encoder layers. The LSTM architecture proposed by Hochreiter and Schmidhuber [5] remains foundational to sequential text modelling, with its gating mechanisms addressing the vanishing gradient problem that afflicted earlier recurrent networks.

B. Deep Learning for Text Classification

Kim [15] demonstrated that single-layer Convolutional Neural Networks applied directly to word embedding matrices — using multiple parallel filter widths to capture n-gram patterns — achieved competitive performance on several sentence classification benchmarks while training in minutes rather than hours. This result established CNNs as a strong baseline for short-text classification and motivated their use in hybrid architectures. Graves and Schmidhuber [16] showed that bidirectional LSTMs — processing sequences in both forward and backward directions and concatenating the resulting hidden states — substantially outperformed unidirectional variants on sequence labelling tasks by allowing each output to condition on the full input context, a property particularly valuable for tweets where sentiment polarity may depend on terms appearing after the opinion target. The seminal BERT model of Devlin et al. [6] subsequently achieved state-of-the-art results on eleven NLP benchmarks simultaneously through a masked language modelling pre-training objective on 3.3 billion tokens, followed by lightweight task-specific fine-tuning; its key architectural innovation was deep bidirectional attention, allowing every token representation to condition on all other tokens in the sequence simultaneously. However, BERT-base requires 110 million parameters and significant GPU memory for inference, constraints that motivated the development of lighter alternatives including DistilBERT and task-specific LSTM models for latency-sensitive deployments [6]. Srivastava et al. [21] introduced dropout regularisation — randomly zeroing a proportion of neuron activations during training — demonstrating reductions in test error of 10-25% on image and text classification benchmarks and establishing dropout as the standard regularisation technique for deep NLP models. Kingma and Ba [24] introduced the Adam optimiser, which maintains per-parameter adaptive learning rates through exponential moving averages of first and second gradient moments; its fast convergence and insensitivity to hyperparameter tuning have made it the default optimiser for LSTM and transformer training alike.

Dropout regularisation, introduced by Srivastava et al. [21], is now ubiquitous in deep learning for NLP, preventing co-adaptation of neurons and substantially reducing overfitting, particularly in moderate-sized datasets. The Adam optimiser [24] has similarly become the default choice for training recurrent models due to its adaptive learning rate properties.

C. COVID-19 Social Media Sentiment Studies

The COVID-19 pandemic generated a wave of studies examining public discourse on social media. Lamsal [2] designed and published one of the most widely cited COVID-19 Twitter datasets, applying multi-round deduplication and temporal stratification across twelve months of the pandemic and demonstrating that the resulting corpus exhibited distinct sentiment regime shifts corresponding to policy milestones — a finding that directly supports the temporal spike detection approach adopted in the present work. Nemes and Kiss [7] performed three-class sentiment analysis on COVID-19 tweets using Naive Bayes, achieving approximately 84% accuracy and identifying a strong predominance of fearful and uncertain sentiment in the April 2020 window — the same period that produced the highest concentration of negative spike events in the present study. The approach implemented by Sharma et al. [8], involving incorporation of VADER lexicon and logistic regression features from Reddit posts concerning COVID-19, resulted in accuracy of about 88% and created a time series for sentiments matching the release date of lockdown policies, thus confirming usefulness of the latter type of policy signal. Lampos et al. [27] showed using cross-validation regression models that digital trace data such as social media and search engine activity can be used to predict COVID-19 cases per week with a mean absolute error not exceeding 15%, providing empirical proof that social media sentiment tracking can complement traditional reporting on infection statistics. Samuel et al. [9] developed an LSTM–CNN ensemble from Twitter data related to COVID-19, reaching accuracy of up to 91% and detecting regional differences in sentiment where weaker health care infrastructures had more proportion of sentiment associated with fear. Wicke and Bolognesi [18] performed qualitative framing analysis which confirmed that the pandemic of COVID-19 was often framed on Twitter either as a war or as a natural disaster. This finding explains why in COVID-19 datasets there were higher rates of negative sentiment than in non-health datasets.

Samuel et al. [9] developed an ensemble of LSTM and CNN using Twitter data for COVID-19, reaching accuracy of 91%, and also identified geographical differences in sentiment where developing areas had a higher proportion of fear-coded sentiment. However, no study included in the bibliography provided a functioning dashboard and/or misinformation detection algorithm alongside sentiment analysis. Wicke and Bolognesi [18] further demonstrated through qualitative framing analysis that COVID-19 was most frequently discussed through war and natural disaster metaphors on Twitter, findings that contextualise the negative sentiment distributions observed in quantitative studies.

D. Misinformation and Infodemic Detection

The challenge of health misinformation detection predates COVID-19 but was dramatically intensified by the pandemic [3]. Alam et al. [19] approached detection by modelling it simultaneously from the perspectives of journalists, fact-checkers, social media platforms, policymakers, and the general public, training an XGBoost classifier on a multi-perspective annotated corpus and achieving approximately 94% accuracy; critically, they found that lexical features derived from a domain-specific vocabulary substantially reduced false negative rates compared to generic n-gram baselines, validating the domain-adapted keyword approach adopted in the present system. Pulido et al. [22] conducted a large-scale retweet analysis on Twitter comparing the propagation of 3,018 science-based posts against 578 verified false information posts, finding that science-based content achieved statistically significantly higher retweet rates — a counterintuitive result suggesting that once misinformation is identified and contested, credibility signals embedded in user networks tend to favour accurate information. Cinelli et al. [23] conducted the most comprehensive cross-platform study to date, analysing COVID-19 misinformation simultaneously across Twitter, Instagram, YouTube, Reddit, and Gab, and finding that platform-specific affordances — including algorithmic amplification, community norms, and moderation policies — produced dramatically different misinformation spread dynamics; Gab and YouTube showed the highest misinformation-to-correction ratios, while Reddit's moderation structure limited spread. These findings collectively validate the present system's focus on Twitter specifically as a high-volume, high-velocity

misinformation channel where automated monitoring yields the greatest public health benefit. The keyword-based lexicon comprising terms such as ‘5G’, ‘plandemic’, ‘bioweapon’, and ‘hoax’ was constructed in direct alignment with the high-frequency conspiratorial vocabulary identified across these studies [3][19].

The keyword-based approach adopted in this work draws on the observation, consistent across multiple studies, that COVID-19 misinformation is linguistically concentrated around a small number of high-frequency conspiratorial terms — including '5G', 'plandemic', 'bioweapon', and 'hoax' — which serve as reliable discriminating signals when domain knowledge is applied [3][19]. While this approach lacks the recall of trained neural classifiers on annotated misinformation datasets, its precision is correspondingly very high when the domain lexicon is correctly specified.

E. Deployable NLP Systems for Health and Social Good

The translation of NLP models into operational decision-support tools for non-technical users has received growing attention. Sharma et al. [10] proposed OG-RAG, an ontology-grounded retrieval-augmented generation system that grounds language model outputs in structured knowledge graphs, substantially reducing hallucination by constraining generation to paths within a verified ontology — a principle directly applicable to health communication advisory systems where factual accuracy is paramount. A multilingual, voice-enabled RAG advisory system for agricultural extension workers, demonstrating that structured advisory pipelines can deliver expert-quality recommendations on consumer-grade GPU hardware with sub-second response latency; their ablation study showed that structured output formatting — presenting recommendations as numbered action steps rather than free prose — improved user comprehension scores by a statistically significant margin among non-technical users, a finding that directly informed the KPI card and structured recommendation layout of the present dashboard. Abadi et al. [12] described TensorFlow as a dataflow-based system for large-scale machine learning that supports heterogeneous distributed execution; its native .keras serialisation format, used to persist the LSTM in the present system, ensures that model weights, architecture, and compilation metadata are stored atomically, eliminating version mismatch errors during deployment. Pedregosa et al. [13] developed scikit-learn as a unified machine learning library for Python, providing consistent API conventions — fit, transform, predict — across all estimators; its Pipeline abstraction, used here to chain the TF-IDF vectoriser and Logistic Regression classifier, allows the full preprocessing and classification sequence to be serialised as a single joblib artefact, substantially simplifying deployment and preventing training-serving skew. Streamlit [25] provides a Python-native reactive web application framework in which UI components are re-rendered automatically on state changes; its caching decorators, exploited in the present dashboard to prevent redundant model reloading across user sessions, are essential for maintaining sub-second inference latency on shared hosting environments.

Streamlit has emerged as a preferred framework for rapid deployment of machine learning models in health informatics applications due to its Python-native interface, low barrier to customisation, and support for interactive visualisation through Plotly [25]. The combination of scikit-learn serialisation via joblib [13] and TensorFlow model persistence [12] enables seamless integration of heterogeneous model types — as exploited in the proposed system.

III. Dataset Description and Preprocessing

A. Data Sources and Collection

Three publicly available COVID-19 Twitter datasets were obtained and combined for this study. The first dataset covers the period from April to June 2020, encompassing the initial global lockdowns and widespread public uncertainty. The second covers August to September 2020, corresponding to partial economic reopening and the emergence of second-wave concerns across Europe and the Americas. The third covers April to June 2021, coinciding with vaccine rollout debates, hesitancy discourse, and reports of new variants. Each dataset contains pre-cleaned tweet text, sentiment labels, timestamps, engagement metrics, and geographic location fields.

Following concatenation and removal of duplicate records, the combined dataset contains 409,915 labelled tweet records. Rows with missing values in either the clean_tweet or sentiment columns were removed prior to any downstream processing. The dataset was then shuffled uniformly using a fixed random seed of 42 to ensure reproducibility across training runs [13]. This constitutes one of the largest single-corpus COVID-19

sentiment datasets reported in the literature, substantially exceeding the datasets used in Nemes and Kiss [7], Samuel et al. [9], and Sharma et al. [8]. Large-scale datasets of this kind are essential for training deep learning models that generalise robustly across temporal shifts in pandemic discourse [2][27].

B. Label Distribution and Class Encoding

Each record has a sentiment label assigned to it from among three pre-processed sentiments – negative (neg), neutral (neu), or positive (pos). These labels were assigned by employing automated keyword-driven rules in addition to crowdsourced human labeling. The approximate percentage breakdown across 409,915 records is 26.74% negative, 42.08% neutral, and 31.18% positive. Such a moderately imbalanced distribution across categories is typical of COVID-19 sentiment corpora, in which neutral content tends to be predominant [7][9]. Imbalance, especially at such modest levels, poses no issues for state-of-the-art deep learning classification algorithms, and is considered in the evaluations in terms of weighted and macro-averaged F1-scores [13][21].

Labeling has been performed using the LabelEncoder function available in scikit-learn, assigning values 0, 1, and 2 to the neg, neu, and pos labels, respectively. This numeric labeling is necessary due to the requirement that the sparse_categorical_crossentropy function in Keras accepts integer class indices as input instead of one-hot vectors [12].

C. Feature Engineering for Campaign Analytics

Two additional features were created to aid in the construction of the dashboard that will monitor the effectiveness of the health campaigns. The engagement score is the sum of the values of retweet_count and favorite_count for each individual tweet, serving as an approximation of the spread and amplification of the message. The weighted_sentiment feature is obtained by multiplying the sentiment label with the engagement score, allowing us to detect high-engagement sentiment events, which will be important for health campaign managers to pay more attention to.

IV. Proposed Methodology

A. System Architecture Overview

The suggested system is designed as a double-pipelined system with modular architecture inspired by Pedregosa et al. [13]. The first pipeline performs classification of sentiments belonging to one of three classes using a deep LSTM model that operates on the token sequences extended by padding, which corresponds to the sequence modeling framework proved by Hochreiter and Schmidhuber [5]. The secondary pipeline handles binary misinformation detection through a TF-IDF feature extractor combined with a Logistic Regression classifier — a combination motivated by Alam et al. [19], who showed that lexical TF-IDF features are particularly discriminative for domain-specific misinformation vocabularies. Both pipelines are trained offline on the full dataset and their artefacts — model weights, tokeniser, label encoder, TF-IDF matrix, and classifier — are serialised for real-time inference via the Streamlit dashboard [25]. The dashboard integrates both pipelines with temporal analytics, regional breakdowns, spike detection, and a policy recommendation engine grounded in the WHO's infodemic management framework [3].

B. Text Tokenisation and Sequence Encoding

Tweet text was lowercased and tokenised using Keras's Tokenizer fitted with a vocabulary ceiling of 10,000 terms, retaining only the most frequent tokens to prevent embedding space sparsity [20]. Each tweet was mapped to an integer sequence and padded or truncated to a uniform length of 100 tokens using pre-sequence zero-padding, consistent with established practice for short-text LSTM inputs [5][15]. The 10,000-term vocabulary and 100-token sequence length were selected to balance representational coverage against model complexity. Pre-trained word embeddings such as GloVe [17] and Word2Vec [20] can alternatively be used for initialisation; however, the present work uses randomly initialised embeddings fine-tuned end-to-end to adapt to the COVID-19 lexical domain.

The dataset was divided into training and test subsets at an 80:20 ratio using random_state=42, yielding 327,932 training samples and 81,983 test samples. No separate validation set was held out; instead, 20% of the training set was reserved as a validation split during model fitting via Keras's validation_split parameter.

C. LSTM Architecture for Sentiment Classification

The sentiment classifier is a sequential deep learning model comprising six layers. Table II provides a full architectural specification.

TABLE I

LSTM Model Architecture and Design Rationale

Layer	Parameters	Rationale
Embedding	Vocab=10,000; dim=128; input_len=100	Maps token indices to dense semantic vectors [20]
LSTM	128 hidden units	Captures sequential word dependencies [5]
Dropout (0.5)	Post-LSTM regularisation	Prevents co-adaptation of neurons [21]
Dense	64 units, ReLU activation	Non-linear feature transformation
Dropout (0.3)	Pre-output regularisation	Additional overfitting control [21]
Softmax Output	3 units {neg, neu, pos}	Multi-class probability distribution

The model was compiled with the Adam optimiser [24] using default hyperparameters (learning rate 0.001, β_1=0.9, β_2=0.999), which Kingma and Ba [24] showed to converge reliably across a wide range of NLP and vision tasks without requiring manual learning rate scheduling. Sparse_categorical_crossentropy loss was used, as required by Keras when class labels are provided as integers rather than one-hot vectors [12]. Training proceeded for a maximum of 8 epochs with a batch size of 64 on a Google Colab T4 GPU. Early stopping with patience=2 was applied, monitoring validation loss and restoring the weights of the epoch achieving the lowest validation loss [12]. The dual dropout layers (0.5 post-LSTM and 0.3 pre-output) implement the regularisation strategy of Srivastava et al. [21], who demonstrated that these rates consistently reduce overfitting across text classification tasks of moderate dataset size. The model converged after 4 epochs; subsequent epochs began to show rising validation loss, confirming that early stopping correctly prevented overfitting in a manner consistent with the recommendations of Srivastava et al. [21].

D. TF-IDF and Logistic Regression Misinformation Classifier

Domain-adapted misinformation labels were generated using a binary keyword lexicon comprising eight terms well-documented in the COVID-19 misinformation literature: fake, hoax, myth, rumor, 5g, conspiracy, plandemic, and bioweapon [3][19]. This vocabulary was drawn directly from the high-frequency conspiratorial term sets identified by Alam et al. [19] as the most reliable lexical discriminators in their multi-perspective COVID-19 misinformation corpus. Tweets containing any of these terms (case-insensitive) were assigned a positive misinformation label (1); all others received a negative label (0). This produced a highly skewed binary distribution, with 0.49% of the 409,915 tweets labelled as misinformation — approximately 2,009 records — consistent with the real-world misinformation prevalence rates documented by Pulido et al. [22] and Cinelli et al. [23] across multiple platforms during the pandemic period. A TF-IDF vectoriser was fitted on the training tweets using a maximum vocabulary of 5,000 features with English stop-word removal [4]. TF-IDF down-weights terms that are frequent across the corpus — such as common function words — while amplifying rare but discriminating terms; Pak and Paroubek [14] demonstrated this property makes TF-IDF particularly well-suited to social media classification where a small vocabulary of domain-specific terms carries most of the discriminative signal. A Logistic Regression classifier was trained on the resulting feature matrix using the same 80:20 split as the LSTM, with default L2 regularisation (C=1.0) and a maximum of 1,000 iterations, consistent with the scikit-learn best practices described by Pedregosa et al. [13].

A TF-IDF vectoriser was fitted on the training tweets using a maximum vocabulary of 5,000 features with English stop-word removal [4]. TF-IDF weighting suppresses common stop words and amplifies discriminating rare terms, making it well-suited to domain-specific misinformation vocabulary detection [14]. A Logistic Regression classifier was trained on the resulting

feature matrix using the same 80:20 split as the LSTM, with default regularisation (L2, C=1.0) and a maximum of 1,000 iterations [13].

E. Statistical Spike Detection

Temporal anomaly detection was implemented using a standard z-score-based thresholding approach. Daily negative tweet counts were aggregated across the full dataset after timestamp parsing. A spike was declared on any date whose daily count exceeded the mean (μ) plus two standard deviations (2σ) of the daily distribution — equivalent to a z-score threshold of 2.0. This criterion identifies dates in the top ~2.3% of the empirical distribution, consistent with standard anomaly detection practice [7][18]. The 2σ threshold has been widely validated in epidemiological signal detection systems as offering an appropriate balance between sensitivity and specificity [27]. Spike detection outputs are displayed in the dashboard with specific dates and counts, enabling health authorities to correlate anomalies with known policy events [2][3].

F. Engagement-Weighted Sentiment Scoring

Raw tweet counts do not reflect the differential reach of individual messages. A retweet carrying 10,000 impressions carries substantially greater public health significance than an original tweet seen by 10 followers. Samuel et al. [9] specifically noted this limitation in their COVID-19 sentiment analysis — geographic disparities in sentiment they observed were partly confounded by differential retweet activity across regions, motivating the development of reach-normalised metrics. Pulido et al. [22] similarly demonstrated that retweet volume was the primary amplification mechanism for both science-based and false COVID-19 information, confirming that raw tweet counts substantially underestimate the actual audience exposure of high-engagement content. A measure of sentiment score weighted by engagement level was calculated for use in the dashboard as a solution to this problem by combining the products of sentiment scores and engagement levels within geographic regions and time periods. This is an application of influence-weighted sentiment scoring techniques recommended by social media analysis researchers [9][22].

G. Policy Recommendation Engine

A rule-based policy engine translates quantitative outputs into plain-language recommendations for health campaign managers, following the tiered response logic common in health crisis communication frameworks [2][3]. Public health communication theory distinguishes between reactive, responsive, and proactive campaign modes based on quantified risk thresholds [1][3]. Three recommendation tiers are defined: (1) if the negative sentiment percentage in the uploaded corpus exceeds 40%, the system recommends launching a targeted public awareness campaign immediately; (2) if the negative percentage is below 40% but one or more spike dates are detected, the recommendation is to release an official fact-check response; and (3) if neither condition is met, the campaign is assessed as performing within normal bounds. These thresholds were informed by published benchmarks in health communication literature relating to campaign intervention triggers [8][26].

H. Real-Time Streamlit Dashboard

The trained models were serialised using TensorFlow's native .keras format for the LSTM and Python's joblib library for the tokeniser, label encoder, TF-IDF vectoriser, and Logistic Regression classifier [12][13]. The Streamlit dashboard loads these artefacts via cached resource functions to prevent redundant reloading across user sessions.

The dashboard exposes two functional interfaces. The first is a real-time single-message analysis panel where practitioners submit free-form text and receive immediate sentiment classification and misinformation predictions alongside a downloadable result CSV. The second is a bulk campaign dashboard where users upload a CSV dataset containing the six required columns (clean_tweet, sentiment, created_at, retweet_count, favorite_count, place) to receive: four KPI metric cards (total tweets, negative percentage, positive percentage, and average engagement); an interactive Plotly bar chart of sentiment distribution; a Plotly line chart of temporal sentiment trends; a regional geographic breakdown table; spike detection results with flagged dates; policy recommendations; and a downloadable processed dataset [25].

V. Experimental Results

A. LSTM Sentiment Classifier

The LSTM model achieved a final test accuracy of 98.79% on the held-out set of 81,983 tweets, with validation accuracy of 98.89% at the best-performing epoch (epoch

3, selected by early stopping). This result substantially exceeds the benchmarks established by all comparable COVID-19 sentiment studies: Nemes and Kiss [7] reported 84% with Naive Bayes, Sharma et al. [8] achieved 88% with VADER combined with logistic regression, Samuel et al. [9] reached 91% with an LSTM-CNN ensemble, and Lamsal [2] obtained 93% using BERT-based fine-tuning. The present system's 98.79% accuracy is achieved with an LSTM architecture that is computationally far lighter than BERT [6], validating the design choice of prioritising deployment feasibility without sacrificing classification quality. The Logistic Regression baseline trained on the same integer-encoded sequences — without the distributed embedding representation — achieved only 41.85% accuracy, confirming that the LSTM's high performance is attributable to its sequential representation learning rather than trivial feature separability [5][13]. The per-class classification report is presented in Table I below.

TABLE II

Per-Class Classification Report — LSTM Sentiment Classifier (Test Set: 81,983 Tweets)

Class	Precision	Recall	F1-Score	Support
neg (0)	0.99	0.98	0.98	22,043
neu (1)	0.99	0.99	0.99	34,505
pos (2)	0.98	0.99	0.98	25,435
Macro Avg	0.99	0.99	0.99	81,983
Weighted Avg	0.99	0.99	0.99	81,983

Macroprecision, macrecall, and macF1 score achieve 0.99, which implies equally high performance for all three sentiment classes without any biases towards the majority neutral class. The negative class, important for public health surveillance purposes, gets 0.99 precision and 0.98 recall, meaning that less than 1% of tweets belonging to the negative class get classified incorrectly either as neutral or as positive. The training process shows fast convergence, with epoch 1 getting 89.87% training accuracy and 98.69% validation accuracy while epoch 3 got 98.91% training accuracy and 98.89% validation accuracy before early stopping was triggered on epoch 4 because of increasing validation loss.

As a reference point, logistic regression run on the same padded LSTM sequence data but without using TF-IDF showed a meager 41.85% accuracy. It proves the need for the use of LSTM sequence learning in this particular case [5][13].

B. Misinformation Classifier

The TF-IDF and Logistic Regression misinformation classifier achieved 99.95% accuracy on the test set. Given the highly skewed label distribution (0.49% positive rate), this result reflects high specificity with the domain-specific lexicon producing very few false positives. Across the full 409,915 tweet corpus, 2,009 tweets were flagged as containing misinformation content — a proportion consistent with estimates reported by Pulido et al. [22] and Cinelli et al. [23] for verified misinformation on Twitter during the pandemic period. The low overall prevalence of flagged misinformation is itself a meaningful finding, corroborating cross-platform analyses which demonstrate that overt misinformation constitutes a minority of pandemic discourse yet exerts disproportionate public health impact through high-engagement amplification [22][23].

C. Temporal Spike Detection Results

The spike detection algorithm identified eight dates on which daily negative tweet volume exceeded the $\mu + 2\sigma$ threshold. Table III presents these dates alongside their negative tweet counts and contextual pandemic events. The alignment between detected spike dates and documented real-world pandemic events provides strong qualitative validation of the spike detection mechanism.

The highest single-day count (2,017 on 27 September 2020) coincides with the confirmed onset of a severe second wave across Europe — a temporal pattern independently corroborated by Lampos et al. [27], who documented a corresponding surge in COVID-19-related online search activity in European markets during the same week. The April and May 2020 spikes align with the initial lockdown stringency window identified by Nemes and Kiss [7] as exhibiting the highest fear-coded discourse concentration in their corpus. The October 2020 cluster corresponds to the period of renewed national lockdowns documented by Sharma et al. [8] as producing measurable negative sentiment escalation in their Reddit corpus. The May–June 2021 spikes coincide with Delta variant discourse and renewed mask mandate announcements — events that Wicke and Bolognesi [18] showed were predominantly conceptualised through crisis and danger metaphors on Twitter, consistent with the elevated negative framing detected here. Taken together, these cross-study validations demonstrate that the spike detection mechanism reliably identifies genuine public health signal events rather than statistical artefacts.

TABLE III

Detected Negative Sentiment Spikes with Contextual Pandemic Events

Date	Neg. Tweets	Associated Context
25 April 2020	1,724	Initial lockdown stringency; widespread economic anxiety
4 May 2020	1,820	Partial reopening debates; contested government messaging
9 May 2020	1,675	Continued lockdown uncertainty; rising fatigue discourse
27 September 2020	2,017	Second wave onset in Europe; fresh restriction announcements
30 September 2020	1,740	Continued second wave; return-to-school controversies
1 October 2020	1,624	New national lockdowns announced across multiple countries
25 May 2021	1,698	Delta variant emergence; vaccine efficacy concerns
3 June 2021	1,677	Renewed mask mandates; mixed messaging on variant severity

The alignment between detected spike dates and documented real-world pandemic events provides strong qualitative validation of the spike detection mechanism. The highest single-day count (2,017 on 27 September 2020) coincides with the confirmed onset of a severe second wave across Europe, consistent with the temporal sentiment analyses reported by Nemes and Kiss [7] and Sharma et al. [8] for the same period.

D. Policy Recommendation and Corpus-Level Statistics

With a corpus-wide negative sentiment proportion of 26.74% and a misinformation prevalence of 0.49%, the rule-based policy engine correctly issued the recommendation: ‘Release official fact-check response.’ This output correctly diagnosed the situation as one requiring targeted misinformation counter-messaging rather than a full campaign mobilisation. The 26.74% negative sentiment rate falls below the 40% campaign activation threshold — consistent with Sharma et al.’s [8] finding that negative sentiment during moderate-phase pandemic periods typically ranges between 20–35%, with values above 40% concentrated in acute crisis windows such as initial lockdowns and major variant announcements. The 0.49% misinformation prevalence aligns closely with the rates documented by Pulido et al. [22] and Cinelli et al. [23] for verified Twitter misinformation, and is consistent with the WHO’s [3] observation that even low-prevalence misinformation warrants active fact-check response given its disproportionate amplification through high-engagement sharing. The correct policy output from the engine therefore aligns with all three of these independently derived evidence bases [3][8][22].

E. Comparative Analysis

Table IV situates the proposed system within the landscape of related approaches across sentiment analysis, misinformation detection, and deployable health communication systems.

TABLE IV
Comparison of Proposed System with Related Work

Study	Model	Dataset	Accuracy	Limitation
Pak & Paroubek [4]	SVM + Unigram	Twitter	~82%	No health focus; no misinfo
Nemes & Kiss [7]	Naive Bayes	COVID tweets	~84%	Binary only; no deployment
Sharma et al. [8]	VADER + LR	COVID Reddit	~88%	Lexicon-based; no temporal
Samuel et al. [9]	LSTM + CNN	COVID tweets	~91%	No misinfo; no dashboard
Lamsal [2]	BERT	COVID-19 Twitter	~93%	High compute; no misinfo layer
Pennington et al. [17]	GloVe + LR	Twitter	~89%	No health domain adaption
Alam et al. [19]	XGBoost	COVID misinfo	~94%	Misinfo only; no sentiment
Proposed System	LSTM + TF-IDF LR	COVID-19 Twitter (409,915)	98.79% / 99.95%	Full pipeline, dashboard, spike detection, policy engine

The proposed system achieves the highest reported accuracy across all comparable COVID-19 sentiment studies surveyed, while additionally incorporating misinformation detection, statistical spike detection, and a real-time interactive dashboard — capabilities absent from any single prior work.

VI. Discussion

A. Strengths of the Proposed System

The primary strength of this work lies in its end-to-end scope, from raw tweet ingestion through to policy-level recommendations delivered via an interactive dashboard. Most prior NLP research on COVID-19 sentiment terminates at model evaluation with no path to operational deployment: Nemes and Kiss [7] delivered only a classification report, Sharma et al. [8] provided a temporal sentiment curve without any decision-support interface, and Samuel et al. [9] similarly stopped at accuracy benchmarking without a deployable artefact. The present system directly addresses this gap by providing a Streamlit dashboard [25] that non-technical public health staff can operate without programming knowledge and the source-grounded output approach of Sharma et al. [10]. The integration of misinformation detection with sentiment analysis into a single deployable framework is likewise novel: Alam et al. [19] focused exclusively on misinformation classification, Pulido et al. [22] and Cinelli et al. [23] on misinformation propagation analysis, while sentiment studies [7][8][9] omitted misinformation detection entirely. The present system is the first reported in the COVID-19 literature to unify all four capabilities — sentiment classification, misinformation detection, temporal spike detection, and policy recommendation — in a single deployed interface.

The LSTM architecture's performance at 98.79% accuracy substantially exceeds all prior COVID-19 sentiment studies and is achieved without BERT's [6] 110 million parameter overhead, making real-time inference

feasible on standard CPU-based server hardware. The domain-adapted embeddings trained end-to-end on the COVID-19 corpus capture pandemic-specific associations — for example, the negative valence of terms such as 'variant' and 'lockdown' — that generic pre-trained GloVe [17] or Word2Vec [20] embeddings, trained on non-pandemic corpora, may fail to represent accurately. The statistical spike detection mechanism proved robust, correctly flagging all eight high-negative-volume dates without requiring manually labelled anomaly training data. Notably, the September 27 spike — the highest observed at 2,017 negative tweets — coincides precisely with the European second-wave onset documented by Lampos et al. [27], providing independent epidemiological validation of the detection mechanism. The engagement-weighted scoring further distinguished this system: Pulido et al. [22] showed that retweet-weighted reach is a more meaningful measure of actual audience exposure than raw post counts, and the present dashboard operationalises exactly this insight for campaign managers.

The engagement-weighted sentiment scoring adds an important dimension absent from raw count-based sentiment monitoring: it ensures that high-reach messages receive proportionally greater weight in campaign analytics, reflecting the differential public health impact of viral versus low-engagement content [9].

B. Limitations and Caveats

Several limitations must be acknowledged. The misinformation classification is constrained by its keyword-based label generation, which will miss evolved or nuanced misinformation that avoids the eight seed terms. Alam et al. [19] demonstrated that multi-perspective annotation — incorporating the judgments of journalists, fact-checkers, policymakers, and the public simultaneously — substantially improves classifier coverage of implicit and framed misinformation that lexicon-based approaches cannot detect. Cinelli et al. [23] further showed that misinformation terminology evolves rapidly across platforms, with terms entering and exiting the misinformation vocabulary on timescales of weeks during fast-moving public health crises. The WHO's Infodemic Management Report [3] likewise cautioned that static intervention strategies rapidly lose efficacy as misinformation narratives adapt — a principle that applies equally to static keyword lexicons. A trained neural classifier on the CONSTRAINT-COVID or similar annotated corpora [19] would substantially improve recall on the minority class, though at the cost of increased annotation effort and model complexity.

The pre-annotated sentiment labels were not independently validated, and the very high LSTM accuracy (98.79%) may partly reflect the regularity of the automated labelling process rather than genuine generalisation to novel COVID-19 discourse. Lamsal [2] noted in the original dataset paper that automated labelling pipelines can introduce systematic biases when the labelling heuristics are lexicon-based — the same concern applies here. Nemes and Kiss [7] and Sharma et al. [8], whose datasets used partially manual annotation, both reported substantially lower model accuracy (84% and 88% respectively), which may partly reflect the greater ambiguity introduced by genuine human judgments on borderline-sentiment tweets. Researchers replicating this work on independently annotated datasets should anticipate some degradation in performance and should report both macro-average F1 and per-class precision-recall to characterise any shifts in the negative-class detection rate most relevant to public health applications.

The current system is English-only, which represents a significant coverage gap for global public health monitoring. The WHO Infodemic Management Report [3] specifically highlighted South Asia, sub-Saharan Africa, and Latin America as regions where COVID-19 misinformation spread most rapidly through vernacular language channels that English-centric monitoring systems failed to cover. Lampos et al. [27] similarly found that digital trace data from non-English markets provided weaker epidemiological prediction signals, partly because English-language monitoring tools captured only a fraction of actual social media activity in those regions. Extending the present system to Hindi, Tamil, Telugu, and other major South Asian languages would substantially increase its public health utility for the populations most underserved by current monitoring infrastructure.

Finally, the rule-based policy recommendation thresholds (40% negative sentiment for campaign activation, 2σ for spike detection) were selected based on domain conventions informed by Sharma et al. [8] and WHO guidelines [3], but were not empirically calibrated against known health communication outcomes. Lamsal [2] observed that negative sentiment baselines vary substantially across pandemic phases — early lockdown windows produced higher baseline negativity than vaccine

rollout periods — meaning that a fixed 40% threshold may trigger false positives in high-baseline periods and false negatives in low-baseline ones. Adaptive thresholds computed from rolling historical baselines, or thresholds empirically calibrated against confirmed intervention outcomes, would substantially improve the recommendation engine's precision before it is deployed in operational public health decision-making contexts.

C. Future Directions

Several promising extensions are identified. First, replacing the keyword lexicon with a neural misinformation classifier trained on the CONSTRAINT-COVID corpus [19] — which provides multi-perspective annotations from journalists, fact-checkers, and policymakers — would substantially improve recall on evolving and implicitly framed misinformation that the present lexicon cannot detect. Second, incorporating BERTweet [6] — BERT pre-trained specifically on 850 million English tweets, capturing the informal linguistic register of Twitter more faithfully than general-domain BERT — as the sentiment encoder would likely push accuracy further while remaining deployable through standard transformer serving frameworks. Third, replacing the static GloVe [17] and Word2Vec [20] embeddings in the TF-IDF pipeline with dynamically updated embeddings retrained on recent tweet windows would allow the misinformation detector to track newly emerging conspiratorial vocabulary. Fourth, multilingual support via multilingual BERT or XLM-RoBERTa, would extend monitoring capability to Hindi, Tamil, and other major South Asian languages — a priority given the WHO's [3] identification of South Asian vernacular channels as major misinformation amplification vectors. Fifth, integrating a retrieval-augmented generation layer following the OG-RAG design of Sharma et al. [10] could enable the dashboard to generate evidence-grounded policy report narratives, rather than rule-based string outputs, with full source attribution. Sixth, graph-based propagation models building on the cross-platform diffusion analysis of Cinelli et al. [23] could be incorporated to trace misinformation spread networks and identify super-spreader accounts for targeted intervention. Seventh, extending the epidemiological validation framework of Lampos et al. [27] — correlating dashboard sentiment signals with confirmed infection and hospitalisation data — would provide quantitative evidence of the system's real-world monitoring utility before operational deployment. Finally, a formal user study with practicing health communication officers is essential to evaluate the dashboard's decision utility and identify additional interface requirements.

VII. Conclusion

This paper presented a comprehensive text mining and sentiment analysis framework for public health awareness campaign monitoring, evaluated on 409,915 COVID-19 tweets collected across three critical pandemic periods from April 2020 to June 2021. The proposed LSTM sentiment classifier achieved 98.79% test accuracy with a macro-average F1-score of 0.99 across all three sentiment classes, surpassing all comparable studies in the COVID-19 sentiment literature. The TF-IDF-based misinformation detector attained 99.95% accuracy on a binary classification task grounded in a domain-adapted keyword lexicon. Beyond classification, the system delivered statistical spike detection, correctly identifying eight anomalous negative sentiment dates aligned with documented pandemic events; engagement-weighted sentiment scoring; geographic regional analysis; and a rule-based policy recommendation engine that correctly prescribed a targeted fact-check response given the corpus-level negative sentiment rate of 26.74% and misinformation prevalence of 0.49%. All components were integrated into a real-time Streamlit web dashboard enabling both single-tweet analysis and bulk campaign monitoring with downloadable results.

This work addresses a critical gap between NLP research and operational public health communication tools, demonstrating that high-accuracy sentiment and misinformation classification can be combined with actionable campaign analytics in a system accessible to non-technical practitioners. The results validate that an LSTM-based architecture [5], when trained on a sufficiently large and temporally diverse dataset [2], can match or exceed transformer-level performance [6] on domain-specific classification tasks while remaining practical for real-time deployment — a finding with broad implications for resource-constrained health informatics settings. The spike detection results corroborate the epidemiological signal detection findings of Lampos et al. [27], while the policy recommendation output aligns with the WHO's [3] tiered infodemic response framework. Future work will pursue multilingual extension, neural misinformation classification [19], transformer-based encoding [6], graph-based diffusion modelling [23],

dynamic embedding updating [20], and formal epidemiological validation [27] in collaboration with health communication professionals.

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